E0123007

```python
import pandas as pd
from sklearn.feature_extraction.text import CountVectorizer
from sklearn.decomposition import LatentDirichletAllocation

documents = [
    "machine learning model for prediction",
    "deep learning and neural networks",
    "artificial intelligence and MAchiine Learning",
    "Football match and term performance",
    "soccer players and Football tatics",
    "Chennaiyin Fc Win the Season 10 Isl Football"
]

vectorizer = CountVectorizer(stop_words='english')
X = vectorizer.fit_transform(documents)

num_topics = 4
lda = LatentDirichletAllocation(n_components=num_topics,
random_state=42)
lda.fit(X)

words = vectorizer.get_feature_names_out()

num_keywords = 5
for topic_idx, topic in enumerate(lda.components_):
    print(f"Topic {topic_idx+1}:")
    top_keywords = [words[i] for i in topic.argsort()[-num_keywords:]]
    print("," .join(top_keywords))
```

```
Topic 1:
term,deep,networks,neural,learning
Topic 2:
performance,machine,prediction,model,learning
Topic 3:
term,intelligence,artificial,machiine,learning
Topic 4:
isl,fc,win,season,football
```